

Robust Control for Frequency Deviation Reduction of a Microgrid With Attached Storage System

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Abstract—This report proposes a Robust MIMO controller synthesized to reduce the frequency deviations in a smart microgrid system which enjoys attached storage. Frequency deviations are associated with load fluctuations and also the renewable energy sources because of their inherent variability. This study considers a microgrid where fossil fuel generators and renewable energy sources are combined with a battery-based storage system. A controller which guarantees Robust stability and performance is synthesized using D-K iteration method. Robustness of the perturbed system is analyzed via investigation of the structured singular values

Index Terms—Robust control, H_∞ control, D-K iteration, μ synthesis, smart grids

I. INTRODUCTION

Recently, remote off-grid Micro-Grid(MG)s have been widely developed especially for rural and distant areas, in which providing electrical energy from the main utility grid is costly and has destructive environmental effects. On the other hand, MG control in an islanded mode is more difficult than the grid-connected mode, because the voltage and frequency regulation of the MGs in grid-connected mode are initially supported by the main utility grid. To improve the efficiency of microgrids and to reduce fossil fuel usage and pollution, renewable energy sources like wind power and solar irradiation may be integrated with traditional power generators such as the diesel ones[1].

Due to the cost effectiveness of wind turbine generation (WTG), it is one of the fastest growing clean power sources. However, since the output power of WTG is proportional to the cube of the (varying) wind speed, it significantly impacts system stability, and can cause large frequency and voltage (F& V) deviations in a micro-grid. In the islanded mode, there are some power generation resources that are responsible for compensation of the fluctuation in load and generation.

Furthermore, high-speed, grid-attached storage systems such as batteries have been proposed for reducing F&V variability. Relatively small amounts of storage can drastically decrease frequency deviation, but only if saturation conditions are avoided by dynamically coordinating storage with other generation sources, using advanced Multi-Input-Multi-Output (MIMO) control approaches. In such circumstances, conventional control approaches are no longer sufficient to constrain these deviations within a small range, and at the same time limit the battery size. More sophisticated robust control algorithms are needed to achieve better performance despite unexpected disturbances and model uncertainties.

The rest of this report is as follows: Section II introduces the plant and its uncertainties. Section III concerns robust

controller design. Finally, Section IV examines the designed robust controller by applying it to the worst perturbed plant.

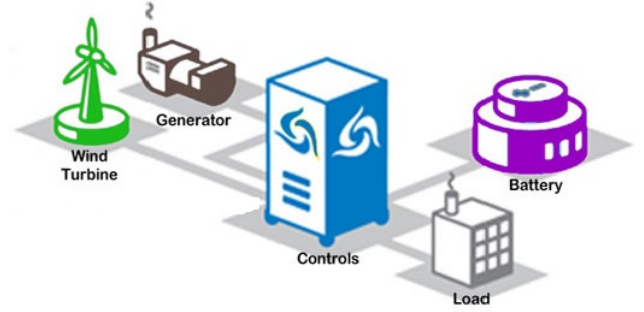


Fig. 1. Structure of microgrid with attached storage system.

II. MODEL STRUCTURE

A typical setup of a microgrid with storage system is shown in Fig. 1. The energy sources include both conventional and renewable generation systems. On the common bus-bar are energy sources, variable loads, and also a battery-based storage system. The essential idea is to increase the usage of renewable energy, and so reduce the fossil fuel consumption, while at the same time rendering the system stable[2]. Here system stability is reflected by incurring only limited system frequency deviations, despite the presence of significant transients. Low frequency load transients are handled by conventional generators (utilizing diesel or natural gas engines as their prime mover). The attached storage system can react much more quickly to load transients, and so it is primarily used for suppressing the high frequency load transients caused by renewable energy sources. In order to maintain the nominal frequency in such a system, more advanced control techniques are required to deliver the system performance requirements.

A. NOMINAL MODEL

The plant is comprised of the interconnection of a WTG, a conventional diesel generator and an attached storage system. The block diagrams of the conventional generator and the storage system are delineated in Figure 2 and 3 respectively. Table 1 shows the parameters of the nominal model. As the

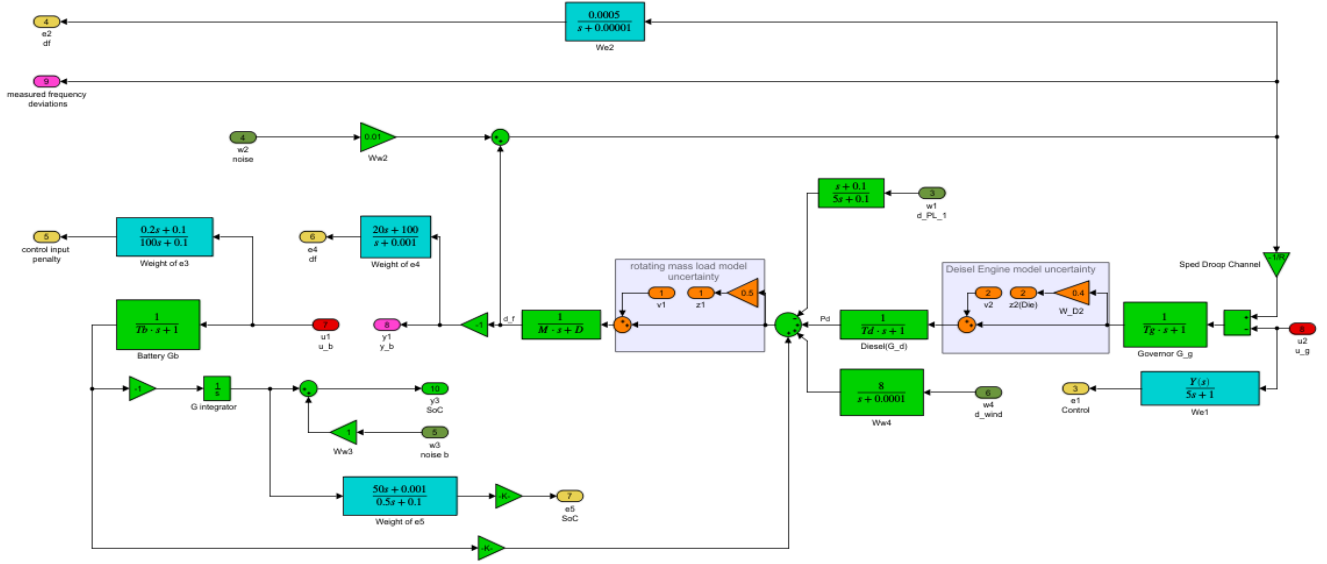


Fig. 5. Plant with model uncertainties for μ synthesis design.

The uncertainties were all modeled as multiplicative SISO perturbations. For μ analysis, the various uncertainties must be extracted from the plant and comprise an uncertainty matrix (Δ) which gets attached to the plant with linear-fractional transformation. The controller connects to the plant in a similar way. Fig. 5 shows this interconnection. The state-space representation (matrices A, B, C, D) of the linearized open-loop plant model (P in Fig. 5) is obtained using the MATLABTM command 'linmod' (applied to Fig. 4), resulting in a system of order 12.

The block-diagonal uncertainty structure, as shown in Fig. 5, is then obtained as an uncertain linear time-invariant object. The Linear Fractional Transformation (LFT) of the linearized uncertain plant (P) and the block diagonal uncertainty structure (Δ) is taken to obtain the weighted, uncertain control design interconnection model. We use the DK-iteration algorithm for -synthesis in Matlab's Robust Control ToolboxTM to design a -optimal robust controller K for our uncertain model. The iterative algorithm combines -synthesis and -analysis to deliver both robustness to uncertainties and optimized performance. The designed robust controller is seen to be of order 12 (same as the design interconnect). This represents a low complexity controller which is easily implemented for real-time operation on modern hardware.

This work develops robust control strategies for both the battery and conventional generation systems, with controllers designed to minimize battery size while at the same time significantly reducing frequency variation, despite variable loads in the microgrid, and the incorporation of a WTG source. The controllers are designed to cope with load transients, WTG output fluctuations, model uncertainties and measurement noise/errors. This study will focus on control of (real) power to reduce frequency deviations.

Conventional generators are slower, and deliver the bulk of the power while coping with large load variations. The resulting closed-loop microgrid system does not control these devices (battery and generator) independently, but rather simultaneously employs the generator to handle large slow load transients, while utilizing the battery to smooth out fast transients. In this way the overall system performance of the microgrid is optimized.

A. SIGNALS' PENALTIES AND WEIGHTING FUNCTIONS

The signals that are subjects to penalty can effectively specify the performance criteria for the controller design optimization process. In order to minimize the system frequency deviation, the first penalized signal is the output. In order to limit excessive usage of the storage system, its SoC signal is penalized as well. In addition, a penalty is always applied on all control signals to limit the control exertion. Here, the control signals act on the diesel engine and the battery.

$$W_{cg} = \frac{(0.2s + 0.1)}{100s + 0.1} \quad (1)$$

$$W_{cb} = \frac{s + 0.0001}{5s + 1} \quad (2)$$

$$(3)$$

Since the generator provides the lower frequency load demands, its weighting function must have larger amplitude in the lower frequency range. The battery fulfills the power demands for the higher frequency range. Its error signal weight must have its low amplitude at the low frequency range, but be significantly high all the way out to very high frequencies.

In other words, during the high frequency range the generator control signal is penalized, since we do not wish

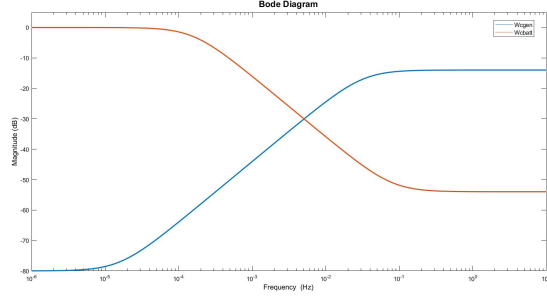


Fig. 6. control inputs' penalty functions

the generator control signal to be active in this range, but rather let the battery deal with small fast transients. On the other hand, the battery control signal weight amplitude is heavily penalized in the low frequency range, since we do not wish the battery to try to handle large slow transients that are better handled by the generator. Through these criteria, the MIMO controller design approach will yield a closed-loop design where each component primarily handles the (frequency) region it is well suited for.

$$W_{w2} = \frac{s + 5e - 4}{s + 1e - 5} \quad (4)$$

$$W_{be} = \frac{20s + 100}{s + 0.001} \quad (5)$$

$$W_{se} = \frac{50s + 0.001}{0.5s + 0.1} \quad (6)$$

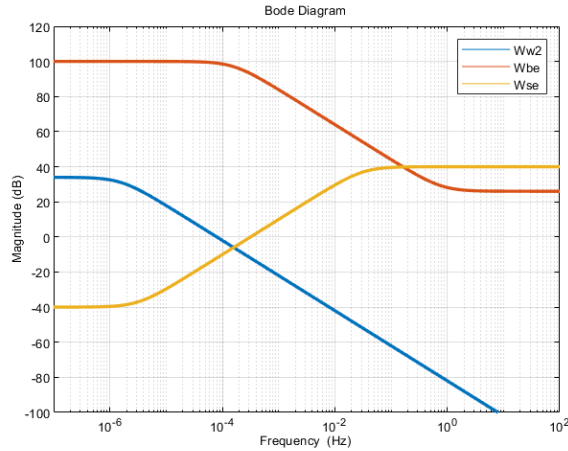


Fig. 7. performance weights

B. H_∞ CONTROLLER

An H_∞ controller that minimizes the upper bound to μ for closed loop nominal system was designed. Fig.6 shows μ analysis to test the robustness of the designed controller. It is clear to see that H_∞ control design lacks robust performance

and robust stability, which means there is a perturbed system that renders the system unstable.

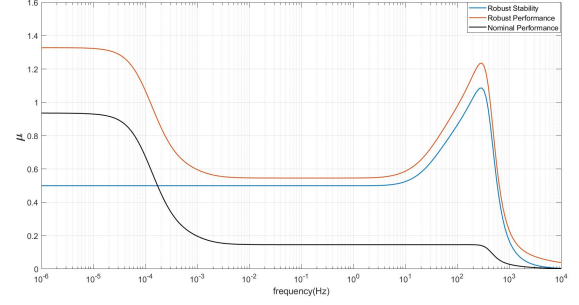


Fig. 8. Closed-loop robustness analysis with K_∞

C. DK ITERATION

Since μ analysis constraint for robust performance ($\mu_\Delta < 1$) was not satisfied, a new controller was designed using DK-iteration. Extracting the first the D-scaling matrices over frequency range then and fitting a 3rd order curve to obtain an approximated transfer function, after two iterations the robust performance (and hence the robust stability too) as well as nominal performance were achieved. DK-iteration and μ analysis results can be found in Fig.5

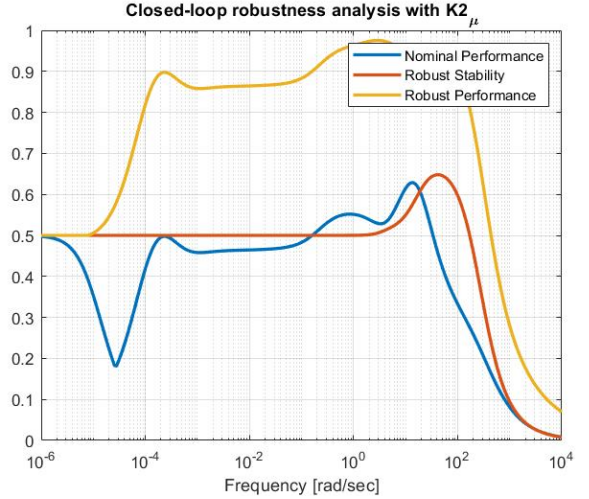


Fig. 9. Closed-loop Robustness analysis with DK-iteration(1) $K1_\mu$

IV. WORST CASE PERTURBATION

In general, for microgrids, Δf should be limited to within 1%, and the recovery time limited to couple of seconds. Otherwise most conventional breakers will trip, with the subsequent possibility of cascade effects. If not, breakers will trip and start disconnecting the loads.

To examine the μ controller's ability to withstand the uncertainties, it is compared to the H_∞ in the worst perturbed model, in presence of the measurement noise, and considerable variations in the powers of load and WTG shown in Fig. 10. A Gaussian measurement noise is added to the model at the w_4 input. Fig. 11 delineates the results. It is evident that the H_∞ controller fails to maintain the frequency constant. The frequency experiences 8Hz of decrease, which is by no means acceptable. On the other hand, when the μ controller is used, the frequency remains fairly constant and the peak of its variations hardly touches 0.2Hz. Therefore, the μ controller succeeds in performing in the worst situation.

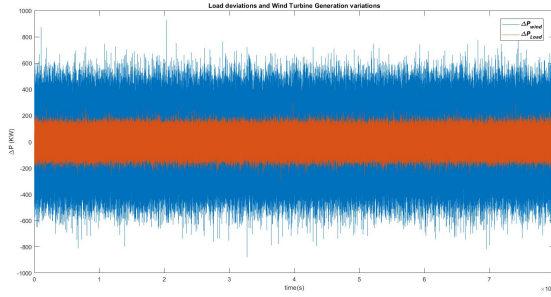


Fig. 10. deviations in ΔP_{wind} and ΔP_{load}

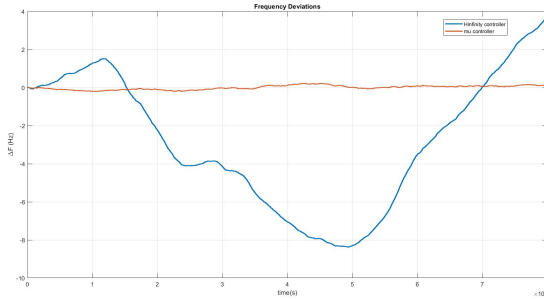


Fig. 11. Comparison of the responses of the system with robust controller and H_∞ controller to harsh ΔP_{wind} and ΔP_{load} for the worst case perturbed plant model

V. CONCLUSION

In this report, an H_∞ controller was designed. From μ analysis data, designing robust controller were motivated. After 2 DK-iterations, the robust stability of system as well as its Robust Performance were guaranteed by the resulting μ controller. The simulation on worst-case scenario of the uncertainties shows the effectiveness of the designed robust

controller. Future works can include the reactive power in the analysis.

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